

PROJECT REPORT DOCUMENTATION

BL Mart Sales Performance ETL Pipeline

Medallion Architecture on the Microsoft Data Platform

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Modelling Approach	Star Schema

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1. Executive Summary

The BL Mart Sales Performance ETL Pipeline is an end to end data engineering solution built on the Microsoft Data Platform. It ingests raw retail sales data, cleans and standardizes it, models it into a dimensional data warehouse, and exposes it for business intelligence reporting in Power BI.

The solution is built on the Medallion Architecture, organizing data into bronze, silver and gold layers of increasing quality and structure, and implements a Star Schema in the Gold layer to support analytical reporting.

This document serves as the formal handover reference for the project. It consolidates the architecture, data model, ETL logic, repository structure and operational notes required for another engineer, analyst or team to understand, maintain and extend the solution.

2. Project Objectives

- Build an end to end ETL pipeline using Microsoft SQL Server and SSIS.
- Clean and standardize raw retail sales data.
- Design and implement a dimensional data warehouse.
- Load dimension and fact tables using stored procedures.
- Create SQL views that answer business questions.
- Build a Power BI dashboard for retail sales analysis.

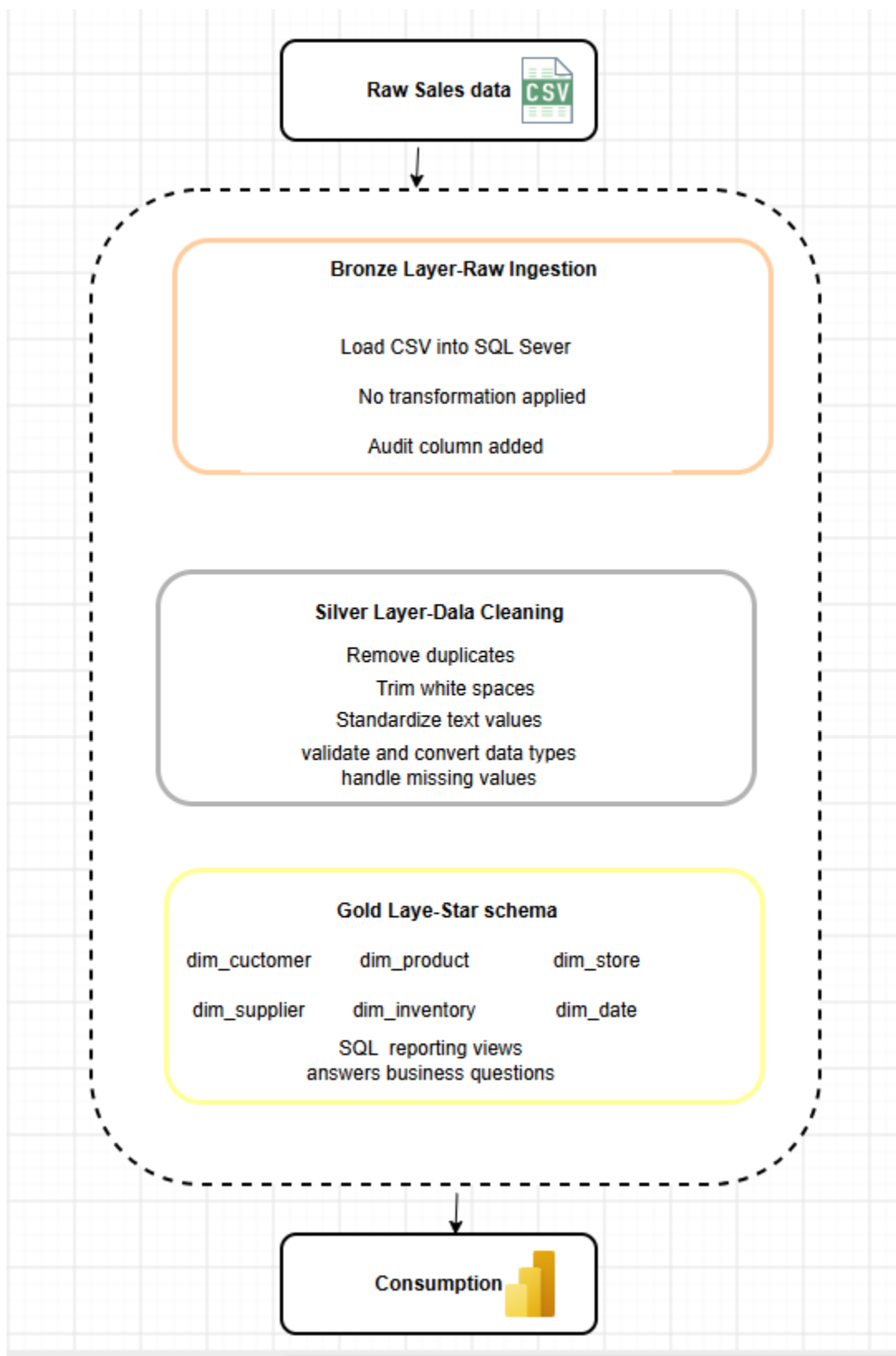
3. Technology Stack

The solution is delivered entirely on the Microsoft Data Platform, supported by Git for version control.

Category	Tooling
Database Engine	Microsoft SQL Server
ETL / Orchestration	SQL Server Integration Services (SSIS)
Database Management	SQL Server Management Studio (SSMS)
Development Environment	Visual Studio
Business Intelligence	Power BI
Version Control	Git, GitHub

4. Solution Architecture

The pipeline follows the Medallion Architecture pattern, progressively refining data from raw ingestion through to a reporting ready dimensional model. Data moves in a single direction, from source files through Bronze, Silver and Gold, before being consumed by Power BI.



Bronze Layer

- Raw data ingestion from source CSV files.
- No transformations are applied.
- Audit columns are added to support lineage.

Silver Layer

- Data cleansing and de duplication.
- Whitespace trimming and text standardization.
- Data type conversion and missing value handling.

Gold Layer

- Star Schema with one fact table and six dimension tables.
 - Surrogate keys generated during load.
 - SQL views built on top of the model to answer business questions.
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5. Data Warehouse Model

The Gold layer implements a Star Schema optimized for analytical querying. Fact_Sales sits at the centre of the model and is surrounded by six conformed dimension tables. Full column level definitions are provided in the companion Data Dictionary referenced in Section 12.

Table	Type	Purpose
Fact_Sales	FACT TABLE	Stores transactional measures and foreign keys to all dimensions. Measures: Transaction Amount, Transaction Discount, Unit Price, Cost Price, Quantity Sold, Line Amount, Stock on Hand.
dim_customer	DIMENSION	Stores customer demographic and loyalty information
dim_product	DIMENSION	Stores product information
dim_store	DIMENSION	Stores retail store information
dim_supplier	DIMENSION	Stores supplier information
dim_inventory	DIMENSION	Stores inventory details.
dim_date	DIMENSION	Stores calendar attributes used for time based analysis

6. ETL Process Detail

Bronze Layer

- Load raw CSV files into SQL Server.
- Preserve original source data without modification.
- Store audit information for every load.

Silver Layer

- Remove duplicate records.
- Trim leading and trailing spaces.
- Standardize text values across fields.
- Validate numeric fields.
- Convert dates to a consistent format.
- Handle null and missing values.

Gold Layer

- Populate dimension tables from cleansed Silver layer data.
- Generate surrogate keys for each dimension.
- Load the Fact Sales table against the conformed dimensions.
- Create reporting views for downstream consumption.

7. Business Questions and Reporting Views

The Gold layer exposes a set of SQL views built specifically to answer recurring retail performance questions.

#	Business Question Answered by a SQL View
1	Top 5 best selling products by revenue
2	Monthly revenue by store
3	Month over month revenue growth
4	Top 10 loyalty customers
5	Customers eligible for a win back campaign
6	Average transaction value by loyalty tier
7	Quantity sold by category and store

8. Repository Structure

The project repository is organized by pipeline layer and deliverable type, as set out below.

```
BL_Mart_Sales_Performance_ETL
├── raw_data
├── database_scripts
│   ├── Bronze
│   ├── Silver
│   ├── Gold
│   └── Views
├── ssis_orchestration
├── powerbi_dashboard
├── documentation
└── README.md
```

9. Environment Setup and Deployment

The steps below describe how to stand up the pipeline in a new environment.

1. Clone the repository to the target machine.
2. Restore the raw CSV files into the raw_data directory.
3. Execute the Bronze layer scripts located in database_scripts/Bronze.
4. Execute the Silver layer scripts located in database_scripts/Silver.
5. Execute the Gold layer scripts located in database_scripts/Gold, followed by the Views scripts.
6. Run the SSIS packages in ssis_orchestration to orchestrate the full pipeline end to end.
7. Open the Power BI file in powerbi_dashboard and refresh the data source against the gold layer.

10. Operational Handover Notes

Run Order

Layers must always be executed in sequence: Bronze, then Silver, then Gold. The Gold layer depends on cleansed Silver data, and reporting views depend on a fully loaded gold layer.

Reprocessing

Should a load need to be repeated, rerun the affected layer and every layer downstream of it, to keep the dimensional model and fact table consistent with the source data.

Ownership

The pipeline, data model and Power BI dashboard were designed, built and are maintained by Mamoraka Letsoalo. Any handover recipient should route pipeline change requests through the documentation and repository structure described in this document.

11. Known Limitations and Future Enhancements

- Orchestration currently relies on SSIS package execution rather than a fully automated scheduler.
- Data quality checks are applied within the silver layer scripts rather than as a standalone validation framework.
- Historical change tracking (slowly changing dimensions) can be extended in a future iteration.
- Source ingestion is currently file based and could be extended to support direct system integration.

12. Related Documents

The following companion document should be read alongside this handover pack:

- BL Mart Sales Performance ETL - Data Dictionary: full column level definitions for the source extract and the gold layer Star Schema.

13. Handover Sign Off

This document represents the complete handover package for the BL Mart Sales Performance ETL Pipeline as at the version stated in Document Control.

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